Real Vessel Data Challenge: A First-of-a-Kind Workshop on Modeling, Insights and Actionable Outcome

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Abstract

The "Real Vessel Data Challenge" initiative is an effort to establish a benchmark for evaluating modeling capabilities among vendors of maritime systems. This workshop challenges participants to analyze real-world vessel data, focusing on tasks such as developing baseline models, estimating fouling effects, and assessing data quality. By providing standardized datasets and encouraging collaborative analysis, the challenge aims to foster innovation and knowledge sharing in the maritime industry. The paper outlines the initiative's objectives, structure, expected outcomes, and milestones, emphasizing its potential to enhance decision-making tools for shipowners and improve methodologies for system vendors.

1. Introduction

Long-term monitoring data has become popular since ISO 19030 was introduced, while even the most reluctant shipowners have installed torque meters after ShaPoLi and practically all are collecting data. Although the ISO is a good start, there is no clear and agreed procedure on how to post-process the data to get meaningful, detailed results which could be transformed into a practical decision-making process, especially in realistic conditions when the data do not have ideal quality and/or distribution. A lot of companies offer long-term monitoring analysis, but there is a wide variety of challenges (i.e. data quality, operational profile, weather effects etc.) and a variety of different needs (i.e. on time hull cleaning, predictive maintenance, impact of ESDs, routing, model predictive control etc.), so it is not necessarily clear how it can tangibly and measurably help shipowners. However, the post-processing of this collected data presents a formidable challenge - one that is not just technical but also multifaceted.

Ship performance teams encounter a variety of issues when dealing with long-term monitoring data: inconsistencies in quality, gaps in datasets, and diverse analytical needs ranging from predictive maintenance to routing decisions. These challenges make it difficult for companies to identify the most effective solutions for their specific requirements. Recognizing this gap, the Real Vessel Data Challenge emerges as a response.

This workshop offers a platform where participants can analyze standardized datasets derived from real vessel operations. The dataset provided includes ISO 19030-compatible data along with ship particulars and event labels from a 76000 tdw Bulk Carrier inviting researchers, engineers, and analysts to explore various aspects of maritime performance. Participants will be tasked with generating baseline models for both clean and current conditions, estimating fouling effects, and critically assessing the quality of the collected data.

Beyond its technical objectives, this challenge fosters a collaborative environment that encourages knowledge sharing among maritime professionals. By providing a forum for diverse methodologies and approaches, it promotes innovation and cross-fertilization of ideas within the industry. The event not only addresses current challenges but also paves the way for future advancements in maritime data analysis.

2. Workshop Objectives

The workshop is designed as an inclusive forum for maritime professionals, researchers, and analysts

from diverse backgrounds to collaborate on pressing challenges in marine data analysis. The objectives of this initiative are broad and multifaceted, aiming not only to address specific research questions but also to foster a sustainable community of practice.

1. Promote Cross-Disciplinary Collaboration: Bring together experts from various fields such as data science, engineering, economics, and policy-making to tackle complex maritime issues collectively.
2. Enhance Research Capacity: Provide researchers with access to unique datasets, and a collaborative platform to conduct high-quality studies on maritime sustainability and safety.
3. Support Open Science Practices: Encourage transparency by sharing datasets, methodologies, and results publicly through an online repository, fostering reproducibility and trust among participants.
4. Foster Innovation in Data Analysis: Offer opportunities for participants to explore new methods and tools for analyzing marine data, addressing gaps in current methodologies.
5. Encourage Peer Learning and Networking: Create a space for knowledge exchange where professionals can share their expertise, learn from others' experiences, and build valuable professional networks.
6. Contribute to Policy Development: Use insights gained from the analysis to inform policy decisions related to maritime safety, environmental regulations, and sustainable practices.
7. Recognize Excellence in Research: Highlight outstanding contributions through awards or recognitions, motivating participants to produce high-quality work.
8. Ensure Ethical Data Usage: Establish clear guidelines for data use to protect confidentiality and ensure responsible sharing among participants.
9. Provide Accessible Resources: Compile a repository of datasets, tools, and detailed analysis reports accessible to future projects, serving as an enduring resource for the field.
10. Encourage Continuous Improvement: Gather feedback post-event to refine subsequent iterations, ensuring each challenge builds on previous successes and lessons learned.
11. Align with Global Initiatives: Integrate findings into broader global efforts in marine conservation and sustainability, contributing to a larger movement towards responsible maritime practices.
12. Facilitate Open Dialogue on Challenges: Create a space for open discussions on encountered obstacles, fostering collective problem-solving and identifying areas needing improvement in methodologies.
13. Ensure Long-Term Accessibility: Archive all materials on a dedicated website for ongoing accessibility and utility post-event.
14. Encourage Feedback for Improvement: Continuously seek participant feedback to enhance the challenge's design and outcomes, ensuring each iteration is better than the last.

By encompassing these objectives, the workshop aims not only to address current challenges in marine data analysis but also to build a lasting community committed to advancing maritime research and sustainability.

3. Challenge Details

For the first installment of this challenge, the choice was made to test the modeling capabilities of the participants for one vessel with relatively good quality of data. More challenging data, and/or more diverse sources that would test the generalization capabilities, will be discussed in future workshops, after evaluation of “simple” capabilities is clarified and established.

Real time (per minute) data of approximately one year will be shared in the form of a csv file. These data will include the following metrics: Draft, Trim, Speed (GPS and Log), Power, RPM, Torque, Turbocharger RPM, Fuel Consumption, and weather data. Along with these data, Daily (noon Report) data will be shared, while ship particulars and sea trials and shop test data will also be shared after anonymization. Cleaning events will also be shared. Based on these data the participants will be required to:

1. Generate baseline models for the vessel in clean condition and current condition: Participants will utilize the datasets provided, to create two baseline models: one representing the vessel in its clean condition and another reflecting its current (latest) operational state. The resulting models will be expressed in the form of Speed-Power Curves, for different drafts, in a predefined (template) format, which will be shared for ease of comparison.
2. Estimate fouling effects in terms of added power and/or speed loss: This involves analyzing how fouling accumulates by assessing the increase in resistance or required power to maintain speed, as well as any potential speed loss. These estimates will be delivered as extra columns to the dataset given
3. Simulate current conditions: Provide estimates of Power and/or Consumption, for periods when these data are missing, when possible.
4. Methodology Comparison: The challenge provides a platform for comparing different modeling approaches, allowing participants to showcase their tools and methodologies effectively. Participants will be allowed to make multiple entries, if they desire, to showcase the capabilities and particularities of different methods / algorithms.
5. Optionally attempt to distinguish hull from propeller fouling: Participants may explore differentiating between fouling effects on the hull and those on the propeller. This optional task adds depth by pushing the methodologies to attempt differentiating between contributors using latent effects.
6. Produce data quality/uncertainty reports: This requires evaluating the consistency, cleanliness, and completeness of the provided data. Participants must identify any gaps or anomalies and quantify uncertainties inherent in their models. Clear documentation here ensures transparency and trust in the analysis results.
7. Open Discussion: Where the participants will be able to comment and/or showcase deeper insights they are willing to share.

By addressing these tasks, participants will share insights into how operational conditions evolve over time and the effectiveness of various mitigation strategies for fouling-related performance issues.

4. Evaluation Criteria / Scoring

To ensure fairness and comparability, the following measurable criteria will be used:

1. Accuracy: Measured using Mean Absolute Percentage Error (MAPE) compared against an undisclosed baseline model generated through Computational Fluid Dynamics (CFD), calibrated using towing tank report.
2. Trend Consistency: Assessed by evaluating how well models maintain consistency over time, particularly during periods of operational changes or environmental variability.
3. Robustness: Tested by exposing models to data quality issues such as outliers and missing values, assessing their ability to provide reliable insights.

Participants will also be encouraged to propose alternative metrics for evaluation during the workshop discussions, ensuring that the challenge remains a dynamic platform for continuous improvement.

5. Milestones Description

Release of Data (May 2025):

- The dataset comprising high-frequency measurements from a 76,000 DWT bulk carrier over a year will be made available to participants via a secure online platform or physical media. (TBA)
- Participants are expected to download the dataset and begin preprocessing it for analysis.

Submission Deadline for Analyses and Reports (December 2025):

- Participants will have from May 2025 until December 2025 to analyze the provided data and submit their findings. This period allows ample time for thorough analysis, model development, and reporting.
- Submissions are expected in a standardized format to ensure consistency across participants and easy comparison.

Compiling Workshop Results (February 2026):

- After the submission deadline, all received analyses and reports will be reviewed and compiled into a comprehensive report by an independent panel of experts (TBD).
- This compilation will summarize key findings, evaluate methodologies used, assess data quality and uncertainty metrics, and provide actionable insights for stakeholders in the maritime industry.

Workshop Presentation (March 2026):

- A workshop will be held in the next HullPIC to present the compiled results and discuss with participants and guests.
- This event will foster collaboration and provide a platform for open discussion on challenges faced during data analysis, potential improvements in methodologies, and future directions.

Publication of Final Report (April 2026):

- The final report will be published online, accessible to all registered participants and industry stakeholders. It will serve as a benchmark document highlighting the achievements of the challenge.
- Based on the level of interest, discussion and outcomes of this first challenge, the next challenges will be defined.

4. Considerations

- Feedback Incorporation: Workshops' feedback will be used to enhance the challenge's effectiveness, ensuring continuous improvement in future iterations. At least one meeting will be arranged, to openly discuss issues or concerns before the submission of analyses.
- Participant Diversity: In order to attract a diverse range of participants from various backgrounds and expertise levels, which will enrich discussions and lead to comprehensive insights, participation and access to the datasets and results will be kept free and open.
- Evaluation Criteria: Clear criteria for evaluating models and reports will ensure fair assessment based on consistent standards. An open forum for the discussion and establishment of fair and unbiased criteria will be established and kept open to all.
- Communication Channels: Establishing forums or discussion boards will facilitate collaboration, allowing participants to share ideas and seek clarification throughout the challenge period. The specific platform(s) (i.e. website, discord server etc.) will be announced soon.

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